

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ 2 SET B**SUMMER SEMESTER, 2024-2025****DURATION: 20 MINUTES****FULL MARKS: 15**

SWE 4805: Software Verification and Validation

Write all the answers on the question paper.

Answer **all 2 (two)** questions. Figures in the right margin indicate full marks of questions.

ID:

1.

Consider the following Alloy model shown in Figure 1 and answer the subsequent question.

10
(CO1)
(PO1)

```
1 sig Customer {}
2
3 sig Product {}
4
5 sig Order {
6     buyer: one Customer,
7     items: some Product,
8     status: one Status
9 }
10
11 abstract sig Status {}
12 sig Placed, Paid, Shipped, Delivered extends Status {}
13
14 sig Payment {
15     order: one Order,
16     amount: one Int
17 }
18
19 sig Delivery {
20     order: one Order,
21     address: one Address,
22     delivered: one Bool
23 }
24
25 sig Address {}
26
27 fact paymentConsistency {
28     all p: Payment | p.order.status in Paid + Shipped + Delivered
29 }
30
31 fact deliveryRule {
32     all d: Delivery | d.order.status in Shipped + Delivered
33 }
34
35 fact nonEmptyOrder {
36     all o: Order | some o.items
37 }
```

Task:

Describe the above Alloy model in natural language. Your answer should include:

- Purpose of each signature and its role in the system, Meaning of each field, and multiplicity constraint and facts

Solution:

An **online shopping system** that manages customers, orders, payments, and deliveries .

- **Customer:** Represents users who place orders in the system.
- **Product:** Represents items available for purchase. Multiple products can be included in a single order.
- **Order:** Represents a purchase made by a customer. Each order:
 - is placed by exactly one customer (`buyer: one Customer`),
 - contains one or more products (`items: some Product`),
 - has a status which represents its lifecycle stage (Placed, Paid, Shipped, Delivered).
- **Status (Abstract):** Represents the state of an order. It is specialized into four stages: Placed, Paid, Shipped, and Delivered.
- **Payment:** Represents the payment associated with an order. Each payment:
 - is linked to exactly one order,
 - has a numerical amount (`Int`).
- **Delivery:** Represents shipment details for an order. Each delivery:
 - is associated with exactly one order,
 - contains a delivery address,
 - stores whether it has been delivered using a boolean flag.
- **Address:** Represents delivery locations for orders.

Explanation of Facts:

- **paymentConsistency:** A payment can only exist for orders that are at least in the *Paid*, *Shipped*, or *Delivered* state. This ensures that payment is not linked to an unprocessed order.
- **deliveryRule:** A delivery is valid only if the associated order is in the *Shipped* or *Delivered* state, ensuring consistency in shipping.
- **nonEmptyOrder:** Every order must contain at least one product, preventing empty purchases.

2.

5
(CO2)
(PO2)

There are seven common myths about Formal Methods in Software Engineering.

- Any **myths you agree with** (and justify why they are true or partially true)
- Any **myth you disagree with** (and explain why it is incorrect)

Your explanation should include reasoning based on Software Verification and Validation concepts.